

Structural and functional evidence supports re-defining mouse higher order visual areas into a single area V2

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eLife Assessment

This paper performs a **valuable** critical reassessment of anatomical and functional data, proposing a reclassification of the mouse visual cortex in which almost all the higher visual areas are consolidated into a single area V2. However, the evidence supporting this unification is **incomplete**, as the key experimental observations that the model attempts to reproduce do not accurately reflect the literature. This study will likely be of interest to neuroscientists focused on the mouse visual cortex and the evolution of cortical organization.

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Abstract

The mouse has become one of the main organisms for studies of the visual system. As a result, there is increased effort to understand universal principles of visual processing by comparing the mouse visual system to that of other species. In primates and other well-studied species including cats and tree shrews, the visual cortex is parcellated into an area V1 and several higher order areas defined by structural and functional differences, and a near complete map of the visual field. In mice, the visual cortex beyond V1 is parcellated into several higher order areas, with less notable structural and functional differences, partial coverage of the visual field, and areal boundaries defined by reversals in progression of the visual field. Notably, recent work in tree shrews and primates has shown that reversals in progression of the visual field can be a hallmark of complex retinotopic mapping within a single visual area. This, and other lines of evidence discussed here, provides a compelling case that the apparent existence of multiple higher order visual areas in the mouse is related to the false assumption of simple retinotopy. Specifically, we use simulations to show that complex retinotopy within a single visual area can recapitulate the appearance of multiple areal borders beyond mouse V1. In addition, we show that many reported differences in functional properties between higher order visual areas can be better explained by retinotopic differences rather than areal identity. Our proposal to reclassify some of the higher order visual areas in the mouse into a single area V2 is not mere semantics because areal definitions influence experimental design and data analysis. Furthermore, such a

reclassification would produce a common set of rules for defining areal boundaries among mammals and would bring the mouse visual system into agreement with evolutionary evidence for a single area V2 in related lineages.

Main text

Beyond V1, the mouse visual cortex has recently been parcellated into many higher order visual areas (**Figure 1A**). These modern area delineations are largely based on anatomical (Wang & Burkhalter, 2007a) and functional (Garrett et al., 2014a; Kalatsky & Stryker, 2003; Zhuang et al., 2017) signatures of a reversal in the progression of the visual field (**Figure 1A**). Such reversals, also known as mirror maps, indicate a new map of the visual field, and therefore a new visual area, under the assumption of simple retinotopic mapping. However, recent work (Sedigh-Sarvestani et al., 2021; Yu, Rowley et al., 2020) has shown that visual field reversals are a hallmark of complex retinotopic mapping within V2 and other higher order visual areas of primates and tree shrews (**Figure 1B**). This prompted us to ask whether some of the higher order areas of the mouse, delineated by reversals, may in fact be a sub-parts of a single area V2 (**Figure 1C**). This would explain the partial visual field coverage of higher order areas in the mouse (**Figure 1A**, Zhuang et al., 2017), which only when combined provide near-full coverage of the visual field.

The parcellation of higher order visual cortex in the mouse has a contentious history (Glickfeld & Olsen, 2017) with different studies parcellating the cortex beyond V1 into between 2 (Rose, Maximilian, 1929) and 16 areas (Zhuang et al., 2017). Many of these studies determine areal boundaries largely based on the projection pattern of V1 terminals in higher order visual cortex. Under the assumption of simple one-to-one mapping, each point in V1 should send projections to only one point in a higher order visual area. Therefore, the presence of multiple segregated projection terminals suggests the presences of multiple higher order areas (Wang and Burkhalter, 2007), but this relationship does not hold if the retinotopic map beyond V1 is complex (Sedigh-Sarvestani et al., 2021; Yu et al., 2020). However, the complexity of retinotopic mapping in the mouse visual cortex has not been investigated, which leads us to consider the areal definitions with skepticism. We are not the first to raise caution regarding this definition. Kaas et al. (1989) argued for a simple higher order cortex with repeated modules within a single area V2. Rosa and Krubitzer (Rosa & Krubitzer, 1999) used comparative anatomical analyses to argue that the presence of multiple visual areas beyond V1 “is not supported by studies of the organization of extrastriate cortex in other mammals, nor by the variability in this organization among extant rodents.” Their words of caution did not impact the next two decades of research into the organization of mouse visual cortex, in part due to the strength of anatomical tracer injections showing multiple visual field reversals beyond V1 (Wang and Burkhalter, 2007), as well as the assumption that retinotopic maps are simple and do not exhibit reversals within a single visual area.

In the past decade, we have gained a better understanding of the diversity of retinotopic transformations in visual areas. Specifically, we have gained structural and functional evidence that visual field reversals can occur in single visual areas when the retinotopic transform within that area is complex. These reversals, and the underlying complex retinotopy, can be explained by wiring minimization principles related to visual field coverage constrained by the anatomy of cortex beyond V1, and have now been reported in tree shrews (Sedigh-Sarvestani et al., 2021), marmosets (Yu et al., 2020), ferrets (Manger et al., 2002a) and squirrels (Gould III, 1984). Therefore, it stands to reason that the mouse visual system could also exhibit complex retinotopic mapping beyond V1, resulting in the false appearance of multiple visual areas, where only a single unified area V2 exists.

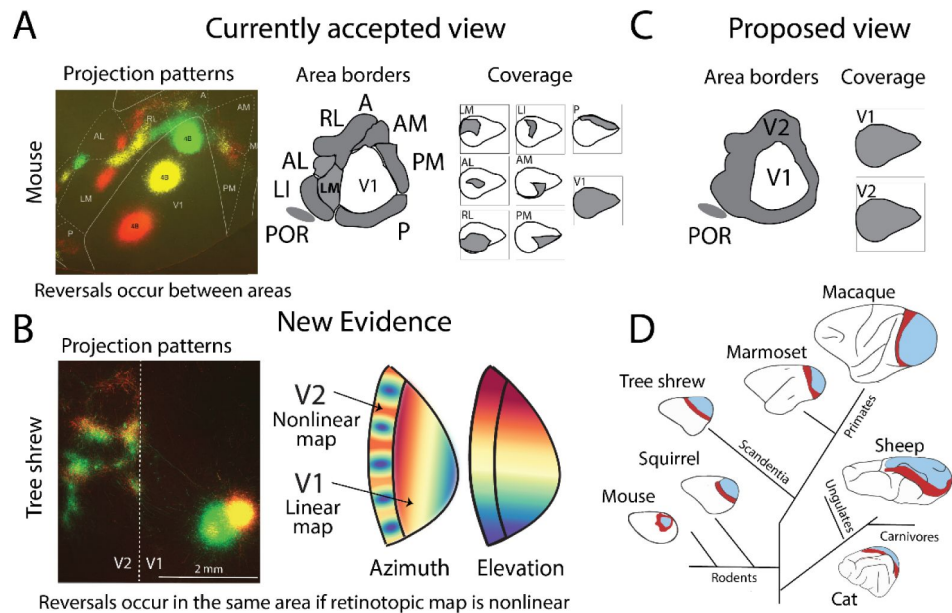


Figure 1

Higher order visual areas in the mouse may be better classified as a single area V2

A) The currently accepted organization of the mouse visual system includes multiple higher order visual areas beyond V1, defined based on reversals in the progression of the visual field, each with partial and biased coverage of the visual field. Projection pattern figure is modified from (Wang & Burkhalter, 2007b) and partial visual field coverage is modified from (Zhuang et al., 2017). **B)** New evidence of visual field reversals within single areas in the tree shrews (Sedigh-Sarvestani et al., 2021), and marmosets (H.-H. Yu et al., 2020) challenge the parcellation of higher order visual cortex in the mouse. **C)** Instead, we propose a new view supports a new view wherein a single area V2, with full coverage of the visual field, borders V1 in the mouse. **D)** This would make the organization of the mouse visual cortex like that in most other mammals, exhibiting an area V1 adjacent to a single area V2. Modified from (Rosa & Krubitzer, 1999)

However, if there is only a single area beyond V1 in the mouse, how can we explain the observed functional differences reported between the higher order visual areas? These include differences, varying smoothly across all higher order areas, in spectral sensitivity (Denman et al., 2018 [DOI](#); Rhim et al., 2017 [DOI](#)), temporal and spatial frequency preferences (Han et al., 2022 [DOI](#); Marshel et al., 2011 [DOI](#)), receptive field size, latency, binocular disparity and several other features. The simplest explanation is that these functional differences are reflective of the different visual field bias of each area (Sedigh-Sarvestani & Fitzpatrick, 2022 [DOI](#)). For instance, we would expect differences in spatiotemporal frequency and receptive field size between area RL and P, simply due to their bias towards lower central and upper peripheral parts of the visual field, which can be cone or rod-driven depending on experimental conditions.

Here we use computational models and a survey of published studies to test the hypothesis that many higher order visual areas beyond V1 can be unified into a single area V2. Specifically, we wondered a) whether published data supports the existence of a complex retinotopic transform beyond V1 and b) what accounts for the observed functional property differences among higher order visual areas as currently defined. We show that established wiring minimization rules that explain the simple retinotopic maps in mouse V1 can be used to explain a complex map beyond V1, belonging to a single secondary area. We also show that the reported functional property differences among higher order areas can be better explained by their visual field bias. More broadly, we find that many differences attributed to distinct areas can be better explained by a single area V2 if one considers two simple facts: visual field reversals are possible under complex retinotopic maps and 2) functional differences exist along the gradient of eccentricity within the same visual area.

The proposed re-unification of mouse higher order visual areas into a single V2 would result in similar rules used to define higher order visual areas across a large array of mammals and marsupials. In addition, the existence of a single area V2 in mice would make the visual cortex of this species consistent with other rodents, and mammals, who exhibit a single area V2 beyond V1 (Rosa & Krubitzer, 1999 [DOI](#)). As it stands, the current definition of visual areas in the mouse makes the organization of visual cortex in this species distinct from nearly all other studied mammals (Figure 1C [DOI](#)), questioning the generalizability and translational potential of findings in the mouse.

Complex retinotopy within a single visual area can recapitulate the appearance of multiple higher order areas in mouse visual cortex

Until improved tracing techniques allowed anatomical mapping of visual field representations beyond V1, the region of visual cortex immediately adjacent to V1 was originally designated as simply V2 based on its cytoarchitecture (Glickfeld & Olsen, 2017 [DOI](#)). Multi-color projection studies (Wang & Burkhalter, 2007b [DOI](#)), however, showed several reversals in progression of the visual field, similar to those observed across the border of V1 and V2 in primates. These reversals were used to delineate area boundaries in the mouse, which produced a series of higher order areas with only partial coverage of the visual field (Garrett et al., 2014b [DOI](#); Zhuang et al., 2017 [DOI](#)).

However, as discussed above, multiple visual field reversals can be a hallmark of complex retinotopic mapping within a single area (Sedigh-Sarvestani et al., 2021 [DOI](#); H.-H. Yu et al., 2020 [DOI](#)). We set out to explain the reversals beyond mouse V1 with an elastic net model of retinotopy, well-established for exploration of map organization in other species (Goodhill & Willshaw, 1990 [DOI](#); Sedigh-Sarvestani et al., 2021 [DOI](#); H.-H. Yu et al., 2020 [DOI](#)). Here, V2 is modeled using the anatomical boundaries of higher order visual cortex, and consists of point “neurons,” each with a receptive field (RF) in the visual field. The objective of the algorithm is to arrange the RFs so they uniformly tile the visual field with the wiring-length constraint of preserving map “smoothness”; e.g., nearby

neurons on the cortex should have nearby RFs in the visual field. The algorithm iteratively optimizes the RFs to reduce the total cost associated with visual field coverage and smoothness until a stable state is reached.

To apply the modeling techniques used in previous papers, we first needed to account for key differences in anatomy of the mouse visual cortex, including the ‘tear-drop’ shaped visual field and the shape of the areas in the cortical sheet (**Figure 2A**). This included modifying the algorithm to weight the interactions between neighbors by the inverse square of their distance in cortical space to account for non-evenly spaced nodes. In addition, we also expected a lower cost associated with the smoothness constraint due to larger receptive fields in the mouse visual system. The mouse’s smaller eyes result in low spatial resolution as the visual field is projected through the pupil onto a smaller retinal surface. In V1 this results in much larger receptive fields and beyond this, allows for bigger steps in the visual field to be taken by adjacent regions of cortex while preserving an overlapping representation. Given this, we expected that a lower weight on the smoothness constraint in our model may result in a better representation of the higher order visual areas’ retinotopy (**Figure 2B-C**).

Figure 2 shows the outcome of an example instance of the model simulating the mouse visual system beyond V1 using 640 neurons. A single elastic net model, constrained by the anatomy of the mouse visual cortex and trained to cover the mouse visual field often (but not always) creates retinotopy with reversals that align to those reported in the mouse. This implies such complex mapping, exhibiting a twisted visual field representation, is the optimal solution to smooth visual field coverage when anatomical constraints of the mouse visual cortex are accounted for. As in previous work, modifying the weights associated with smoothness and isotropy results in internal reversals, seen in alternating field sign (Sereno et al., 1994) (**Figure 2C-E**). The precise layout of these reversals varies somewhat with the initial random seed, but always includes multiple reversals in the upper and lower visual field representations. We note that variation also exists in the reported retinotopy of the mouse between animals (Garrett et al., 2014a; Zhuang et al., 2017). There may also be additional constraints in the mouse visual cortex that are not accounted for in our model that could result in a more robust outcome. Further work is needed to create a more reliable model, but the model as it stands clearly demonstrates the feasibility of a single mouse V2 to account for the complex retinotopic mapping, multiple visual field reversals, and the appearance of multiple visual areas under the false assumption of simple retinotopy.

The twisted nature of the single visual field covering higher order visual cortex can also be shown in **Figure 2D**, where azimuth and elevation axes are plotted against each other in X and Y coordinates at the base, and the progression along the visual field as one moves around V1 on the cortex, is plotted in the Z-axis. This complex retinotopy generates, from a single continuous area, reversals that approximately match mouse higher order visual areas (**Figure 2E**). In fact, we can use the same model to simulate multi-color tracer injections into V1 (**Figure 2F**), and show several modular projection zones, which would appear to be multiple distinct representations of the visual field without knowledge of the underlying complex transform between V1 and V2. Together, our simulation data provides strong evidence in support of the unification hypothesis and suggests that previously reported multiple higher visual areas with partial retinotopic maps can be more parsimoniously explained with a single area with a full, but twisted and complex, representation of the visual field. However, multiple differences between these areas along anatomical and functional lines have also been reported. Next, we will review and assess if a single V2 area with a warped retinotopy could account for these differences.

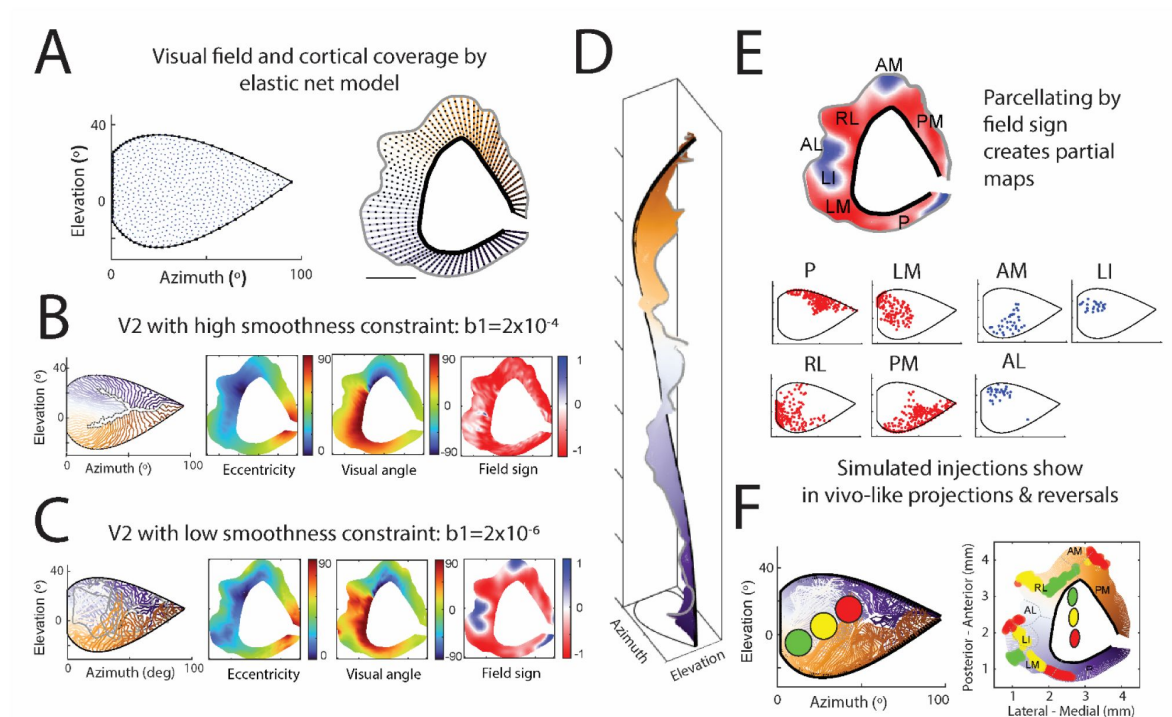


Figure 2

Retinotopic models show emergence of a single complex map of visual space beyond mouse V1

We extended the elastic net model from previous work (Sedigh-Sarvestani et al., 2020) to constrain a model of V2 to the unique structure of the mouse. **A**) Borders of cortical space were traced from (Zhuang et al., 2017) and a teardrop shaped visual field was used to approximate the full visual field covered by mouse V1. **B-C**) Modifying the constraints to smoothness and isotropy result in internal reversals, seen in alternating field sign (Sereno et al., 1994). **D**) Visualization of the visual field, azimuth & elevation planes, covered by normal lines between the V1 and higher order visual area border as in right of A. These lines are stacked on the Z-axis, and color coded, going from the caudal border of V1 in purple around the circumference of V1 to the medial edge in orange. **E**) This complex retinotopy generates, from a single continuous area, reversals that approximately match mouse higher order visual areas, showing that previously reported retinotopic maps in mouse higher order visual areas are parsimonious with a single area covering a full, but twisted, visual field. **F**) Matching points in the visual field with their locations in this single area, creates similar patterns to what has been shown in tracer studies (Wang & Burkhalter, 2007a), compare 1A.

Reported functional property differences among higher order areas are better explained by retinotopic bias and opsin gradients

Functional feature differences are one of the main metrics used to justify areal definitions in the higher order visual system of primates and cats. For instance, area MT in primates exhibits sensitivity to global motion direction that is absent in area V2, even in identical regions of the visual field. Such hierarchical and parallel processing of overlapping parts of the visual field is thought to represent an efficient strategy to build selectivity to complex input features (Felleman & Van Essen, 1991 [↗](#)).

While there is growing evidence of functional diversity within mouse visual cortex, most observations of diverse functional properties report relatively small and continuous changes in tuning across multiple areas, rather than robust and sharp differences between higher order areas as observed in primates (Figure 3 [↗](#)). For instance, binocular disparity tuning (Chiomia et al., 2019 [↗](#)), color tuning (Aihara et al., 2017 [↗](#)), and coherent motion processing (Sit & Goard, 2020 [↗](#)) have all recently been shown to vary across the elevation axis of retinotopy (Figure 3 A [↗](#)). These trends are consistent with the ethological relevance of visual field locations for a low-lying animal like the mouse (Qiu et al., 2021a [↗](#)). Furthermore, the areas categorized into the dorsal and ventral stream of the mouse cortex also exhibit robust biases for the lower and upper visual field (Figure 3B [↗](#)). This further supports the major effect of retinotopy on the structure and function of the mouse visual cortex.

Functional property differences in higher order visual areas have also been reported to vary along the azimuth axis of retinotopy. For instance, receptive field size, response latency, and the degree of phase-locking to drifting gratings vary smoothly across higher order visual areas, along azimuth (Siegle et al., 2021 [↗](#)). The effect of optogenetic inhibition of secondary visual areas also seems to exhibit a retinotopic bias, with inhibition of lateral areas producing little effect on a contrast detection task, and inhibition of the medial areas producing a robust effect (Goldbach et al., 2021 [↗](#); Jin & Glickfeld, 2020 [↗](#)). However, this effect may not arise from simple stimulus-location effects but may be due to inherent differences in the contribution of different regions of the visual field to specific perceptual tasks. Finally, encoding of various behavioral task-related features was shown to vary across higher order visual areas of the mouse (Minderer et al., 2019 [↗](#)), but as the authors point out, this variation is agnostic to higher order areal boundaries and instead follows the underlying topography smoothly (Figure 3C [↗](#)). In fact, the only functional discontinuities in the mouse posterior cortex were found at the major anatomical borders between V1, parietal, and retrosplenial areas.

Such observations of smooth functional property differences across mouse higher visual cortex, coupled with the fact that retinotopic locations within a single visual area often exhibit smoothly varying functional differences (Groen et al., 2021 [↗](#); Kanwisher, 2001 [↗](#); Sedigh-Sarvestani & Fitzpatrick, 2022 [↗](#); H. H. Yu & Rosa, 2014 [↗](#)), leads us to a more parsimonious explanation: Functional features differences in mouse higher order cortex can be simply explained by the underlying retinotopic bias across areas. Testing this hypothesis requires careful measurements of visual field location and functional property differences in V1 and higher order visual areas. Such data sets are sparse but growing. To test our hypothesis with existing data, we reanalyzed a dataset from Han et al. (Han et al., 2022 [↗](#)) and showed that some of the reported differences in spatial frequency (SF) and temporal frequency (TF) tuning can be accounted for by a smooth function of eccentricity and elevation (Figure 3 [↗](#)). The relationship of tuning with eccentricity is unexpectedly inverted from what has been reported in V1 but is nonetheless still smooth.

In primate vision research ‘eccentricity’ is used to mean distance in visual degrees from the center of gaze, which is a useful metric for animals with a fovea and robust overrepresentation of the central few degrees of vision. Many functional properties scale with eccentricity within visual

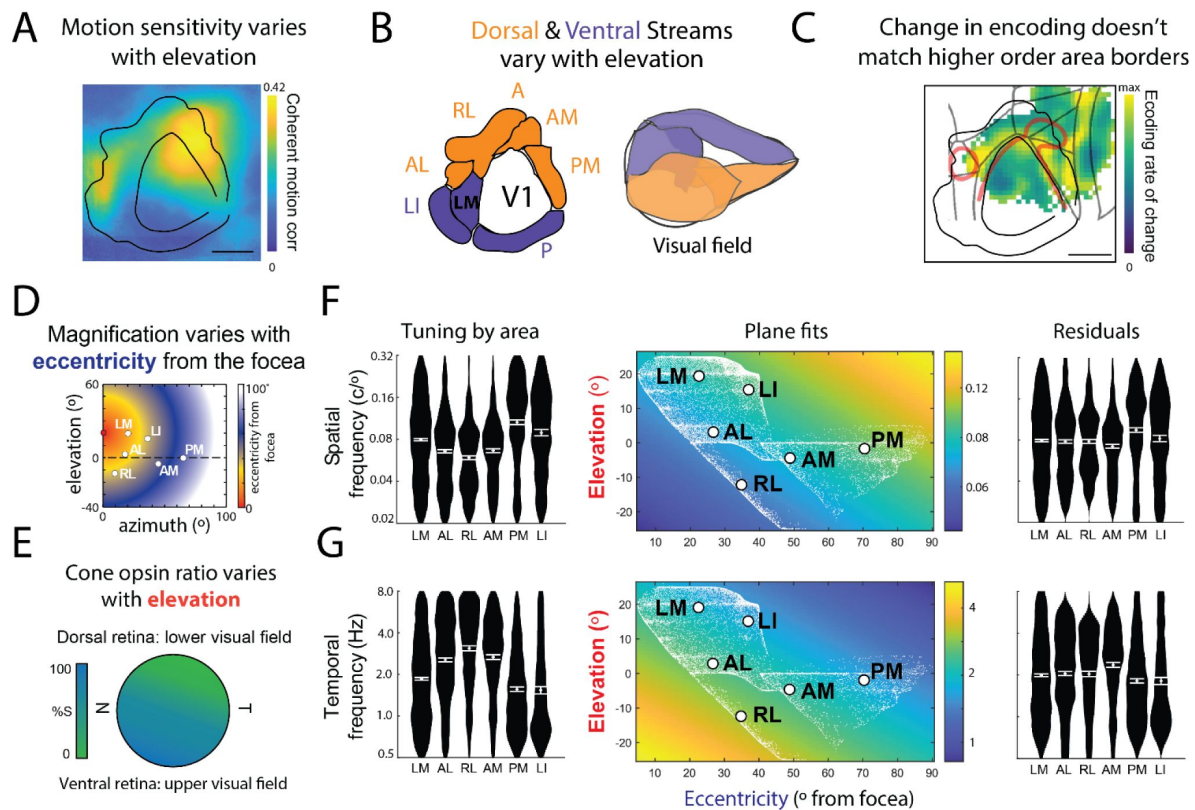


Figure 3

Retinotopy accounts for functional variations between higher order areas in mouse visual cortex

A) Map of motion sensitivity adapted from (Sit & Goard, 2020) showing higher sensitivity in the lower field representation of V1 and higher order visual areas. **B**) Dorsal and Ventral streams in the mouse containing multiple higher order visual areas have been suggested due to differences in function, however these maps closely align onto elevation axis of retinotopy. Figure made using data from (Zhuang et al., 2017) **C**) Behavioral encoding as a function of cortical space shows large change along the V1 border, but not between higher order visual areas (Minderer et al., 2019). Scale bar 1mm. **D**) Schematic of eccentricity from fovea with area estimates for the data used in F&G. **E**) Schematic of retinal cone opsin gradient following (Demb & Singer, 2015). **F**) Data from (Han et al., 2022). Tuning to spatial frequency (SF) of bandpass cells in each area (left), plane fit of SF to area elevation and eccentricity from fovea (middle) and the residuals (right) after subtracting a plane fit. **G**) Same for temporal frequency (TF). Area tuning violin plots show the distribution of tuning within areas with white dots for the mean and lines for 95% confidence interval.

areas, making it a very important parameter to control for in foveal animals such as primates (Yu et al., 2015). However, in the afoveate mouse, ‘center of gaze’ is difficult to control for and typically elevation above eye level and azimuth relative to a normal through the eye are used.

While this is a reasonable approach for an animal without a fovea, it has been reported that a central region of the upper binocular visual field of the mouse is slightly overrepresented in V1 and seems to be behaviorally significant. This region was named the ‘focea’ (van Beest et al., 2021). Due to strong expectations that functional response properties scale with distance from this point, we will assess functional properties as a function of distance from the focea. Taking the focea as 20 degrees above resting eye level on the vertical meridian (0 degrees azimuth), ‘eccentricity’ (Figure 3D) is simply the Euclidean distance from this point in degrees of visual field.

There is an additional effect across retinotopy in the mouse that must be considered. The mouse retina has a gradient of cone opsins (Figure 3E) such that S-cones, sensitive to UV light, are overrepresented on the ventral surface (the upper visual field) and give way to M-cones, sensitive to green light, on the dorsal surface (the lower visual field). When a typical LCD monitor is used, M-cones but not S-cones, are partially activated (Rhim et al., 2021). This means that retina in the upper and lower visual fields are in a rod-dominated scotopic and mixed cone- and rod-mediated mesopic regime respectively. This retinal gradient is known to be preserved in the visual cortex (Rhim et al., 2017) and is thought to play an important role in predator evasion (Qiu et al., 2021b). Thus, control for the opsin gradient is crucial for accurate measurement, reporting, and comparative analysis of functional properties in mice. Specifically, the use of typical LCD monitors results in primacy of scotopic vision in the ventral retina, producing higher spatial and lower temporal frequency tuning in the upper visual field.

In Han et al. (2022) an LCD screen was used. The observed trend, albeit small, towards higher SF tuning in the areas with predominantly upper field representations (Figure 3F) suggests that the cone opsin gradient may be a factor. To account for the effects of both eccentricity from the focea and the cone opsin gradient across elevation, we fit a plane to the peak tuning as a function of these variables. The dataset includes retinotopic mapping from a clockwise circling visual stimulus, revealing reversals that were used to register data to a common map. Here, to get a more fine-grained estimate of the retinotopic locations, we further registered these to the Zhuang et al. (2017) maps for azimuth and elevation. To account for the cells that had reported tuning at the extreme edges of the range, the data was fit using expectation maximization algorithm as noisy data varying around a mean centered on a plane. The proportion of censored values was thus able to contribute to the plane fits without introducing a bias by removing them. We also excluded area P as this area wasn’t separated from POR in Han et al. (2022). In Figure 3F-G, we show that subtracting the expected peak tuning of each cell calculated from a plane fit through eccentricity and elevation significantly reduces the functional property differences among higher order areas. This suggests that a large portion of the measured differences in spatial and temporal frequency tuning in higher order areas can be explained by their bias towards different regions of the visual field.

We note that changes in spatial and temporal frequency, varying smoothly with eccentricity, have been observed in visual areas of various species including primates (Groen et al., 2021; Kanwisher, 2001; Sedigh-Sarvestani & Fitzpatrick, 2022; H. H. Yu & Rosa, 2014). However, in the case of spatial frequency, the relationship we observe in the mouse secondary visual cortex seems to be inverted from that observed in primates (Yu et al., 2015): preference for higher spatial frequencies is seen in the peripheral, rather than central vision. This has been reported in higher order visual areas of the mouse by several independent groups (Han et al., 2022; Marshel et al., 2011). However, such an inverse relationship between SF preference and eccentricity does not impact our argument that functional feature differences smoothly vary across multiple areas, rather than sharply in between higher order areas – suggesting that functional feature

differences observed in the mouse do not lend support to the delineation of multiple areas. Further work is needed to explore why the neuronal preferences for spatial frequency and eccentricity seem to co-vary with opposite trends in the visual cortex of rodents and primates. Based on this data, we believe the functional feature differences observed across the mouse higher order visual system can be explained more parsimoniously by variations in retinotopy rather than areal identity. This limits the use of functional feature diversity in justifying areal definitions. To rely on functional feature differences to assess areal identity, the confound of retinotopy must be removed. This is typically done by reporting functional feature differences across areas, in overlapping parts of the visual field representation of each area. The studies mentioned above do not carry out this important control, largely due to the fact the higher order visual areas in the mouse do not have overlapping representations of the visual field (Zhuang et al., 2017 [↗](#)). This lack of visual field overlap across areas, is in itself, suggestive of a lack of hierarchical multi-area processing in the mouse. The lack of visual field overlap combined with the observed smooth changes in functional feature tuning strongly supports the re-classification of multiple higher order visual areas into a single area V2 (**Figure 1C** [↗](#)). Such an area would exhibit the observed functional feature differences in different parts of its visual field map, as has been observed in primates and cats. Such differences ultimately arise due to the presence of different types of ethologically relevant information in distinct parts of the visual field (Sedigh-Sarvestani et al. 2022).

Reported anatomical evidence for multiple higher order visual areas can be explained by complex retinotopy in a single area V2

Efforts to understand the organization of the mouse visual system were largely based on prior anatomical and functional measurements in primates. Early parcellations of the mouse visual cortex were made based on cytoarchitectural features including cell density and laminar distribution - leading to the delineation of two subcomponents of a single area V2 (Caviness, 1975 [↗](#)). Future work showed boundaries in labeling of histological markers such as SMI-32 and m2ChR labeling, but such changes mostly delineated area LM/AL (Wang et al., 2011 [↗](#)) and seemed to be correlated with the representation of the lower visual field. We now know, based on work in both mice and primates that different parts of the retinotopic map of a single visual area can exhibit different genetic and molecular markers, different cell types, different cell density, and robustly different anterograde and retrograde connections to other areas (Morimoto et al., 2021 [↗](#); Mundinano et al., 2019 [↗](#); Palmer & Rosa, 2006 [↗](#); H.-H. Yu et al., 2015 [↗](#)). Therefore, observations of histological differences in higher order visual cortex, on their own, cannot be used to delineate areal boundaries.

Later parcellations of the mouse visual cortex relied, in addition to cytoarchitecture, on the pattern of anterograde connections from V1 to higher order visual cortex, as well as the pattern of callosal connections from contralateral V1 into ipsilateral higher order visual cortex. Combining these projection patterns revealed a complex set of modules in higher order visual cortex - each assumed to be a separate area (**Figure 1A** [↗](#)). We now know that both the modular mirror-map organization of anterograde projections from V1, and callosal projections from contralateral V1, can be explained by non-retinotopic mapping within a single visual area (**Figure 1B** [↗](#)), as in tree shrews (Sedigh-Sarvestani et al., 2021 [↗](#)). The definition of a single visual area despite the appearance of multiple mirror maps, or reversals, relies on the existence of a single mapping of a visual field - with a complex transformation occurring between the retina and V2. This is precisely the case in the visual cortex of the mouse, where multiple modules (currently defined areas) combine to form a single, complex, representation of the visual field. Each module, when considered alone, represents only a subsection of the visual field, with little overlap between modules (**Figure 1A** [↗](#)). Therefore, given what we now know about complex retinotopic transforms, observations of patchy or modular organization beyond V1 cannot be used to delineate multiple areal boundaries.

In addition to cytoarchitecture and connection patterns, a third anatomical feature used to delineate the primate visual cortical hierarchy is pairwise asymmetries in feedforward and feedback connection densities and laminar distributions. Specifically, feedforward inputs tend to terminate in primate layer 4 whereas feedback inputs tend to terminate in superficial and deep layers, but not in layer 4. In the rat and mouse, the laminar distribution is less distinct between feedforward and feedback inputs, but D'Souza et al. (2022) [discovered](#) a pattern specific to the mouse, focused on differences between Layer 1 and Layer 2-4. Feedforward inputs tend to terminate in Layers 2-4, but not in Layer 1, and feedback inputs tend to terminate in Layer 1, and less so in Layers 2-4. One can quantify this relationship using an optical density ratio (ODR) of labeling density in Layer 1, vs. all layers. An analysis based on such laminar differences easily distinguished V1 as providing feedforward input to higher order areas, with ODR values less than 0.5. Higher order areas were also easily distinguished as providing feedback inputs to V1, as evidenced by ODR values significantly greater than 0.5. However, the connections between higher order visual areas exhibited ODR values near 0.5, suggesting that these areas form neither feedforward nor feedback connections to each other, but are rather at the same level of the hierarchy. This lends further support to the unification of these higher order areas into a single V2, producing a simple 2-layer shallow hierarchy in the mouse visual cortex consisting of V1, V2, and perhaps the outlier area POR as a third layer.

Complex retinotopy beyond V1 is a common feature of secondary visual cortex in multiple species

The visual cortex beyond V1 of several species contains modular organization with multiple reversals – as in the mouse visual cortex. These include tree shrews as previously mentioned, but also ferrets (Manger et al., 2002a [discovered](#)), and macaques (Roe & Ts'o, 1995b [discovered](#), 1995a [discovered](#)). In fact, if one considers the visible portion of V1 and V2 in these species, it bears a striking resemblance to the analogous regions in the mouse visual cortex (**Figure 4** [discovered](#)), with all species exhibiting simple retinotopy in V1, and complex and modular retinotopy immediately beyond V1. However, despite the presence of multiple reversals as in the mouse, the current classification of visual cortex (**Figure 4** [discovered](#), blue box) includes a single area V2 in both the tree shrew and macaque visual cortex. In the ferret visual cortex, two areas are defined: area 18 and area 19. This is contrast to the current classification of the mouse visual cortex, which includes multiple areas beyond V1, parcellated based on visual field reversals (**Figure 4A** [discovered](#)). The disparity in the rules used to delineate the visual cortex of mice, vs other species is highlighted when reversals are used to delineate areas in the visual cortex of tree shrews, macaques, and ferrets (**Figure 4** [discovered](#), “Borders using reversals”). In each case, many areas would be defined beyond V1.

Given this disparity, one could argue that the current classification of the mouse visual cortex can be left intact, and the organization of the visual cortex in other species should be updated to reflect the reversal rule. This would result in common areal delineations across species. However, we believe the existing data strongly supports a single area V2 bordering V1 in all species, including mice. Below we discuss the rational for defining a single, or at most two, visual areas bordering V1 in tree shrews, ferrets and macaques.

In tree shrews (**Figure 4B** [discovered](#)), the visual cortex adjacent to V1 exhibits a modular organization, with retinotopic stripes that alternate between representations of relatively more central and relatively more peripheral regions of the visual field. In the middle of each stripe lies a visual field reversal. However, a unified V2 is defined in the tree shrew which subsumes these reversals. The rational for this delineation include cytoarchitecture, projections from V1, and cross-hemispheric projections (Kaas et al., 1972 [discovered](#); Sesma et al., 1984 [discovered](#); Wong & Kaas, 2009 [discovered](#)). In addition, despite the modular appearance of the retinotopy map, it has been shown that there is a single visual field represented in V2 with no repeated coverage (Sedigh-Sarvestani et al., 2021 [discovered](#)). Therefore, just as in mouse visual cortex, complex retinotopy gives the appearance of multiple visual field

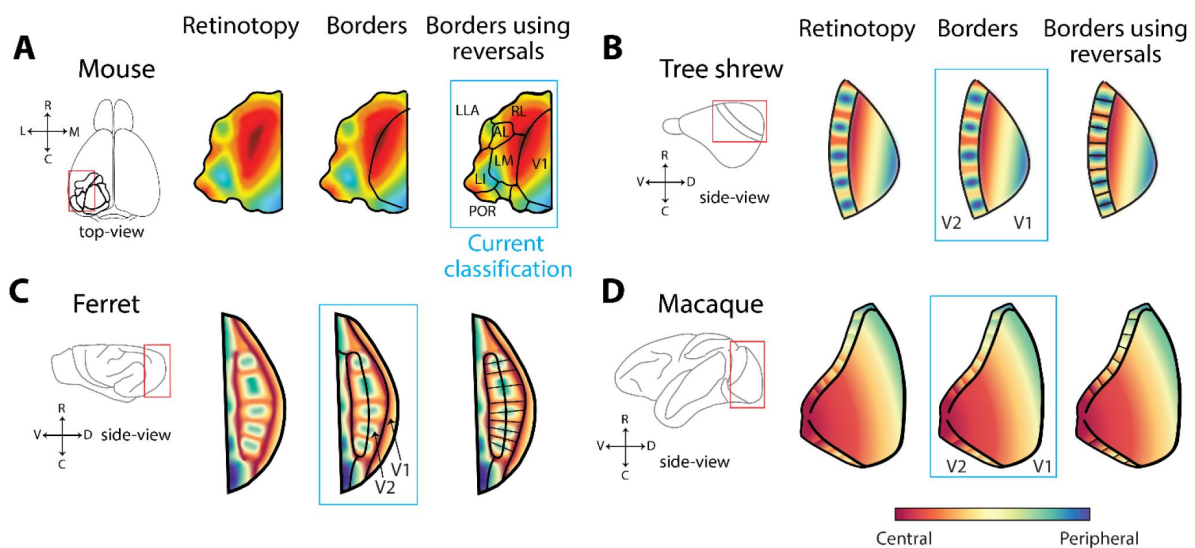


Figure 4

Complex retinotopy beyond V1 is a common feature of secondary visual cortex in multiple species

A) Retinotopy maps in mouse V1, schematics created based on data from (Zhuang et al., 2017). Currently accepted area delineations shown in blue box. **B)** Retinotopic maps in tree shrew visual cortex. Data from (Sedigh-Sarvestani et al., 2021). **C)** Retinotopic maps in ferret visual cortex. Data from (Manger et al., 2002a). **D)** Schematic of retinotopic maps in visible portion of macaque V1 and V2. Data from (Roe & Ts'o, 1995b; Shipp & Zeki, 2002b). Color maps are different for each species and are simply meant to indicate relatively more central or peripheral regions of the visual field.

representations where only a single representation exists. Furthermore, functional property differences observed in V2 stripes can largely be attributed to underlying retinotopic differences (Sedigh-Sarvestani et al., 2021 [↗](#)).

In ferrets (**Figure 4C** [↗](#)), a similar modular representation was observed beyond V1, with repeated modules of crossings between more central and peripheral regions of the visual field. Within each central or peripheral module lies a visual field reversal. However, these reversals are not used to delineate multiple areas bordering V1. Instead, an area 18 (V2) and an area 19 (V3) are defined, based on cytoarchitecture, anatomy, and functional properties (Dell et al., 2019 [↗](#); Manger et al., 2002a [↗](#), 2002b [↗](#)). As in tree shrews, despite the presence of multiple reversals, there is a single visual field representation across area 18 and 19, with each module covering a small portion of the visual field (Manger et al., 2002b [↗](#)). Similar observations have been made in the cat secondary visual cortex (Boyd & Matsubara, 1994 [↗](#)).

A series of elegant papers in the 1980s and 1990s showed a modular, striped, organization in macaque V2 (Hubel & Livingstone, 1987 [↗](#); Roe & Ts'o, 1995b [↗](#); Shipp & Zeki, 2002a [↗](#)). The stripes exhibited distinct functional feature preferences, including for color, motion, and horizontal disparity. The underlying retinotopic map has been less clearly defined, in part due to the lack of optical imaging measurements in macaque V2. Published reports based on electrode penetrations across V2 indicate multiple reversals between relatively more foveal or peripheral regions of the visual field (Roe & Ts'o, 1995b [↗](#); Shipp & Zeki, 2002b [↗](#)). Though these reversals were interpreted as multiple representations of the visual field, we suspect they form a single unified representation similar to tree shrews and ferrets. Nonetheless, despite the appearance of multiple representation of the visual field, a single area V2 was defined in macaques – in part because of the clear modular and repeated nature of functional property differences across the length of V2.

In summary, there appears to be a common complex organization beyond V1, exhibiting modularity, in several evolutionarily distant species. Cytoarchitecture, anatomy, and functional measurements support defining these modules into a single area V2.

Discussion

Here we have made a case, using simulations and a critical re-analysis of published data, that many of the higher order visual areas in the mouse are actually sub-parts of a single area V2. We have shown that multiple visual field reversals, on their own, cannot be used to delineate areal boundaries. This is because complex mapping of the visual field can produce multiple reversals within a single area. We have also shown that many of the functional property differences reported between higher order areas are better explained by retinotopic bias among areas and known differences in functional feature preferences across the retinotopic map of a single visual area. We believe this evidence strongly supports redefining mouse higher order visual areas into a single area V2. What remains, is a visual system with distinct properties across its visual field - aligned with ethological needs and behaviors. This new definition produces a common set of rules for defining areal boundaries among mammals: the presence of a full visual field, homogenous cytoarchitecture and smooth functional properties.

We emphasize that our proposal goes beyond ever-green arguments on ‘what is a visual area?’. Ideally, we would be able to define areal boundaries using automated, model-based, approaches (Shimaoka et al., 2024 [↗](#)). However, we currently lack the cross-species data required to develop such models with meaningful accuracy. Until then, the least we can do is to ensure that we use the same rules to define visual areas across species, regardless of what the rules are per se. In the case

of visual cortex beyond V1, applying such universal definitions for areal boundaries would bring mice into agreement with evolutionary evidence for a single area V2 in other nearby lineages (Rosa and Krubitzer, 1999 [↗](#)).

However, there remains some evidence against the hypothesis of a single V2 in mouse visual cortex. Harris et al. (Harris et al., 2019 [↗](#)) used the laminar density of projection patterns between different cortical and subcortical regions to give classification scores to the different higher order visual areas and found significant different hierarchy scores among these areas. However, they note that the hierarchy was much shallower than expected, particularly between these areas. Future work should assess whether hierarchy scores calculated from different regions of mouse V1 would exhibit a similar shallow network simply due to retinotopic biases in functional feature properties and laminar connection densities.

We note, again, that our argument is not novel. Rosa and Krubitzer made the same argument and plea in 1999, based on the presence of a single area V2 in a large range of mammals and non-mouse rodents (**Figure 1D** [↗](#)). We build on their argument armed with the knowledge of complex retinotopy that can explain the apparent differences in anatomy and function. More importantly, we believe the distinction of a single area into multiple areas has misled the field in both experimental design and interpretation of data collected from mouse visual cortex. Therefore, a redefinition is not merely a matter of semantics but rather necessary for improved studies of the visual system.

We believe in the value and urgent need for cross-species comparisons in neuroscience. This manuscript presents an effort on our part to align the mouse with other mammals. With a single area V2 definition, the mouse visual cortex is no longer an outlier in the evolutionary tree. This means we can continue to rely on this species to gain knowledge about the mammalian visual system. Otherwise, we should think carefully about whether what we learn in the mouse can be applied to any other mammal.

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References

- Aihara S., Yoshida T., Hashimoto T., Ohki K (2017) **Color Representation Is Retinotopically Biased but Locally Intermingled in Mouse V1** *Frontiers in Neural Circuits* **11** <https://doi.org/10.3389/fncir.2017.00022> | Google Scholar
- Boyd J., Matsubara J (1994) **Tangential organization of callosal connectivity in the cat's visual cortex** *Journal of Comparative Neurology* **347**:197–210 <https://doi.org/10.1002/cne.903470205> | Google Scholar
- Caviness V. S. (1975) **Architectonic map of neocortex of the normal mouse** *J. Comp. Neurol* **164**:247–263 <https://doi.org/10.1002/cne.901640207> | Google Scholar
- Chioma A. L., Bonhoeffer T., Hübener M (2019) **Area-Specific Mapping of Binocular Disparity across Mouse Visual Cortex** *Current Biology* **29**:2954–2960 <https://doi.org/10.1016/j.cub.2019.07.037> | Google Scholar
- Dell L. A., Innocenti G. M., Hilgetag C. C., Manger P. R (2019) **Cortical and thalamic connectivity of occipital visual cortical areas 17, 18, 19, and 21 of the domestic ferret (*Mustela putorius furo*)** *Journal of Comparative Neurology* **527**:1293–1314 <https://doi.org/10.1002/cne.24631> | Google Scholar
- Demb J. B., Singer J. H (2015) **Functional Circuitry of the Retina** *Annual Review of Vision Science* **1**:263–289 <https://doi.org/10.1146/annurev-vision-082114-035334> | Google Scholar
- Denman D. J., Luviano J. A., Ollerenshaw D. R., Cross S., Williams D., Buice M. A., Olsen S. R., Reid R. C (2018) **Mouse color and wavelength-specific luminance contrast sensitivity are non-uniform across visual space** *eLife* **7** <https://doi.org/10.7554/eLife.31209> | Google Scholar
- D'Souza R. D., Wang Q., Ji W., Meier A. M., Kennedy H., Knoblauch K., Burkhalter A (2022) **Hierarchical and nonhierarchical features of the mouse visual cortical network** *Nature Communications* **13**:503 <https://doi.org/10.1038/s41467-022-28035-y> | Google Scholar
- Dyck R. H., Cynader M. S (1993) **An interdigitated columnar mosaic of cytochrome oxidase, zinc, and neurotransmitter-related molecules in cat and monkey visual cortex** *Proceedings of the National Academy of Sciences of the United States of America* **90**:9066–9069 <https://doi.org/10.1073/pnas.90.19.9066> | Google Scholar
- Felleman D. J., Van Essen D. C. (1991) **Distributed hierarchical processing in the primate cerebral cortex** *Cerebral Cortex* **1**:1–47 <https://doi.org/10.1093/cercor/1.1.1-a> | Google Scholar
- Garrett M. E., Nauhaus I., Marshel J. H., Callaway E. M (2014a) **Topography and areal organization of mouse visual cortex** *J. Neurosci* **34**:12587–12600 <https://doi.org/10.1523/JNEUROSCI.1124-14.2014> | Google Scholar
- Garrett M. E., Nauhaus I., Marshel J. H., Callaway E. M (2014b) **Topography and Areal Organization of Mouse Visual Cortex** *Journal of Neuroscience* **34**:12587–12600 <https://doi.org/10.1523/JNEUROSCI.1124-14.2014> | Google Scholar

- Glickfeld L. L., Olsen S. R (2017) **Higher-Order Areas of the Mouse Visual Cortex** *Annu Rev Vis Sci* **3**:251–273 <https://doi.org/10.1146/annurev-vision-102016-061331> | Google Scholar
- Goldbach H. C., Akitake B., Leedy C. E., Histed M. H (2021) **Performance in even a simple perceptual task depends on mouse secondary visual areas** *eLife* **10**:e62156 <https://doi.org/10.7554/eLife.62156> | Google Scholar
- Goodhill G. J., Willshaw D. J (1990) **Application of the elastic net algorithm to the formation of ocular dominance stripes** *Network: Computation in Neural Systems* **1**:41–59 https://doi.org/10.1088/0954-898X_1_1_004 | Google Scholar
- Gould H. J (1984) **Interhemispheric connections of the visual cortex in the grey squirrel (*Sciurus carolinensis*)** *Journal of Comparative Neurology* **223**:259–301 <https://doi.org/10.1002/cne.902230209> | Google Scholar
- Groen I. I. A., Dekker T. M., Knapen T., Silson E. H (2021) **Visuospatial coding as ubiquitous scaffolding for human cognition** *Trends in Cognitive Sciences* <https://doi.org/10.1016/j.tics.2021.10.011> | Google Scholar
- Han X., Vermaercke B., Bonin V (2022) **Diversity of spatiotemporal coding reveals specialized visual processing streams in the mouse cortex** *Nature Communications* **13**:3249 <https://doi.org/10.1038/s41467-022-29656-z> | Google Scholar
- Harris J. A., Mihalas S., Hirokawa K. E., Whitesell J. D., Choi H., Bernard A., Bohn P., Caldejon S., Casal L., Cho A., Feiner A., Feng D., Gaudreault N., Gerfen C. R., Graddis N., Groblewski P. A., Henry A. M., Ho A., Howard R., ..., Zeng H (2019) **Hierarchical organization of cortical and thalamic connectivity** *Nature* **575**:195–202 <https://doi.org/10.1038/s41586-019-1716-z> | Google Scholar
- Hubel D. H., Livingstone M. S (1987) **Segregation of form, color, and stereopsis in primate area 18** *Journal of Neuroscience* **7**:3378–3415 <https://doi.org/10.1523/jneurosci.07-11-03378.1987> | Google Scholar
- Jin M., Glickfeld L. L (2020) **Mouse higher visual areas provide both distributed and specialized contributions to visually guided behaviors** *Current Biology : CB* **30**:4682–4692 <https://doi.org/10.1016/j.cub.2020.09.015> | Google Scholar
- Kaas J. H., Hall W. C., Killackey H., Diamond I. T (1972) **Visual cortex of the tree shrew (*Tupaia glis*): Architectonic subdivisions and representations of the visual field** *Brain Research* **42**:491–496 [https://doi.org/10.1016/0006-8993\(72\)90548-3](https://doi.org/10.1016/0006-8993(72)90548-3) | Google Scholar
- Kaas J. H., Krubitzer L. A., Johanson K (1989) **Cortical connections of areas 17 (V-I) and 18 (V-II) of squirrels** *The Journal of Comparative Neurology* **281** <https://doi.org/10.1002/cne.902810308> | Google Scholar
- Kalatsky V. A., Stryker M. P (2003) **New paradigm for optical imaging: Temporally encoded maps of intrinsic signal** *Neuron* **38**:529–545 [https://doi.org/10.1016/s0896-6273\(03\)00286-1](https://doi.org/10.1016/s0896-6273(03)00286-1) | Google Scholar
- Kanwisher N (2001) **Faces and places: Of central (and peripheral) interest** *Nature Neuroscience* **4**:455–456 <https://doi.org/10.1038/87399> | Google Scholar
- Manger P. R., Kiper D., Masiello I., Murillo L., Tettoni L., Hunyadi Z., Innocenti G. M (2002a) **The representation of the visual field in three extrastriate areas of the ferret (*Mustela***

ptorius) and the relationship of retinotopy and field boundaries to callosal connectivity *Cerebral Cortex* (New York, N.Y.: 1991) **12**:423–437 <https://doi.org/10.1093/cercor/12.4.423> | [Google Scholar](#)

Manger P. R., Kiper D., Masiello I., Murillo L., Tettoni L., Hunyadi Z., Innocenti G. M (2002b) **The representation of the visual field in three extrastriate areas of the ferret (Mustela putorius) and the relationship of retinotopy and field boundaries to callosal connectivity** *Cerebral Cortex* **12**:423–437 [Google Scholar](#)

Marshel J. H., Garrett M. E., Nauhaus I., Callaway E. M (2011) **Functional specialization of seven mouse visual cortical areas** *Neuron* **72**:1040–1054 <https://doi.org/10.1016/j.neuron.2011.12.004> | [Google Scholar](#)

Minderer M., Brown K. D., Harvey C. D (2019) **The Spatial Structure of Neural Encoding in Mouse Posterior Cortex during Navigation** *Neuron* **102**:232–248 <https://doi.org/10.1016/j.neuron.2019.01.029> | [Google Scholar](#)

Morimoto M. M., Uchishiba E., Saleem A. B (2021) **Organization of feedback projections to mouse primary visual cortex** *iScience* **24**:102450 <https://doi.org/10.1016/j.isci.2021.102450> | [Google Scholar](#)

Mundinano I.-C., Kwan W. C., Bourne J. A (2019) **Retinotopic specializations of cortical and thalamic inputs to area MT** *Proceedings of the National Academy of Sciences* **116**:23326–23331 <https://doi.org/10.1073/pnas.1909799116> | [Google Scholar](#)

Palmer S. M., Rosa M. G. P (2006) **A distinct anatomical network of cortical areas for analysis of motion in far peripheral vision** *European Journal of Neuroscience* **24**:2389–2405 <https://doi.org/10.1111/j.1460-9568.2006.05113.x> | [Google Scholar](#)

Qiu Y., Zhao Z., Klindt D., Kautzky M., Szatko K. P., Schaeffel F., Rifai K., Franke K., Busse L., Euler T (2021a) **Natural environment statistics in the upper and lower visual field are reflected in mouse retinal specializations** *Current Biology* **31**:3233–3247 <https://doi.org/10.1016/j.cub.2021.05.017> | [Google Scholar](#)

Qiu Y., Zhao Z., Klindt D., Kautzky M., Szatko K. P., Schaeffel F., Rifai K., Franke K., Busse L., Euler T (2021b) **Natural environment statistics in the upper and lower visual field are reflected in mouse retinal specializations** *Current Biology* **31**:3233–3247 <https://doi.org/10.1016/j.cub.2021.05.017> | [Google Scholar](#)

Rhim I., Coello-Reyes G., Ko H.-K., Nauhaus I (2017) **Maps of cone opsin input to mouse V1 and higher visual areas** *J. Neurophysiol* **117**:1674–1682 <https://doi.org/10.1152/jn.00849.2016> | [Google Scholar](#)

Rhim I., Coello-Reyes G., Nauhaus I (2021) **Variations in photoreceptor throughput to mouse visual cortex and the unique effects on tuning** *Sci. Rep* **11**:11937 <https://doi.org/10.1038/s41598-021-90650-4> | [Google Scholar](#)

Roe A. W., Ts'o D. Y (1995a) **Visual Topography Functional Stripes** *The Journal of Neuroscience* **15**:3689–3715 [Google Scholar](#)

- Roe A. W., Ts'o D. Y (1995b) **Visual topography in primate V2: Multiple representation across functional stripes** *Journal of Neuroscience* **15**:3689–3715 <https://doi.org/10.1523/jneurosci.15-05-03689.1995> | [Google Scholar](#)
- Rosa M. G., Krubitzer L. A (1999) **The evolution of visual cortex: Where is V2?** *Trends Neurosci* **22**:242–248 [https://doi.org/10.1016/s0166-2236\(99\)01398-3](https://doi.org/10.1016/s0166-2236(99)01398-3) | [Google Scholar](#)
- Rose Maximillian (1929) **Cytoarchitektonischer atlas der Groshirnrinde der Maus** *J Psychol Neurol* **40**:1–32 [Google Scholar](#)
- Sedigh-Sarvestani M., Fitzpatrick D. (2022) **What and Where: Location-Dependent Feature Sensitivity as a Canonical Organizing Principle of the Visual System** *Frontiers in Neural Circuits* **16**:834876 <https://doi.org/10.3389/fncir.2022.834876> | [Google Scholar](#)
- Sedigh-Sarvestani M., Lee K. S., Satterfield R., Shultz N., Fitzpatrick D (2020) **A sinusoidal transform of the visual field in cortical area V2** *bioRxiv* <https://doi.org/10.1101/2020.12.08.416651> | [Google Scholar](#)
- Sedigh-Sarvestani M., Lee K.-S., Jaepel J., Satterfield R., Shultz N., Fitzpatrick D (2021) **A sinusoidal transformation of the visual field is the basis for periodic maps in area V2** *Neuron* **109**:4068–4079 <https://doi.org/10.1016/j.neuron.2021.09.053> | [Google Scholar](#)
- Sereno M. I., McDonald C. T., Allman J. M (1994) **Analysis of retinotopic maps in extrastriate cortex** *Cereb. Cortex* **4**:601–620 <https://doi.org/10.1093/cercor/4.6.601> | [Google Scholar](#)
- Sesma M. a, Casagrande V. a, Kaas J. H. (1984) **Cortical connections of area 17 in tree shrews** *The Journal of Comparative Neurology* **230**:337–351 <https://doi.org/10.1002/cne.902300303> | [Google Scholar](#)
- Shimaoka D., Wong Y. T., Rosa M. G. P., Price N. S. C (2024) **Naturalistic movies and encoding analysis define areal borders in marmoset third-tier visual cortex** *Progress in Neurobiology* **240**:102657 <https://doi.org/10.1016/j.pneurobio.2024.102657> | [Google Scholar](#)
- Shipp S., Zeki S (2002a) **The functional organization of area V2, I: specializatoiin across layers** *Visual Neuroscience* **19**:211–231 <https://doi.org/10.1017/s0952523802191176> | [Google Scholar](#)
- Shipp S., Zeki S (2002b) **The functional organization of area V2, II: The impact of stripes on visual topography** *Visual Neuroscience* **19**:211–231 <https://doi.org/10.1017/s0952523802191176> | [Google Scholar](#)
- Siegle J. H., Jia X., Durand S., Gale S., Bennett C., Graddis N., Heller G., Ramirez T. K., Choi H., Luviano J. A., Groblewski P. A., Ahmed R., Arkhipov A., Bernard A., Billeh Y. N., Brown D., Buice M. A., Cain N., Caldejon S., ..., Koch C (2021) **Survey of spiking in the mouse visual system reveals functional hierarchy** *Nature* **592**:86–92 <https://doi.org/10.1038/s41586-020-03171-x> | [Google Scholar](#)
- Sit K. K., Goard M. J (2020) **Distributed and retinotopically asymmetric processing of coherent motion in mouse visual cortex** *Nature Communications* **11**:3565 <https://doi.org/10.1038/s41467-020-17283-5> | [Google Scholar](#)
- van Beest E. H., Mukherjee S., Kirchberger L., Schnabel U. H., van der Togt C., Teeuwen R. R. M., Barsegyan A., Meyer A. F., Poort J., Roelfsema P. R., Self M. W. (2021) **Mouse visual cortex**

contains a region of enhanced spatial resolution *Nature Communications* **12**:4029 <https://doi.org/10.1038/s41467-021-24311-5> | [Google Scholar](#)

Wang Q., Burkhalter A (2007a) **Area map of mouse visual cortex** *J. Comp. Neurol* **502**:339–357 <https://doi.org/10.1002/cne.21286> | [Google Scholar](#)

Wang Q., Burkhalter A (2007b) **Area map of mouse visual cortex** *Journal of Comparative Neurology* **502**:339–357 <https://doi.org/10.1002/cne.21286> | [Google Scholar](#)

Wang Q., Gao E., Burkhalter A (2011) **Gateways of ventral and dorsal streams in mouse visual cortex** *J. Neurosci* **31**:1905–1918 <https://doi.org/10.1523/JNEUROSCI.3488-10.2011> | [Google Scholar](#)

Wong P., Kaas J. H (2009) **Architectonic subdivisions of neocortex in the tree shrew (*Tupaia belangeri*)** *Anatomical Record* **292**:994–1027 <https://doi.org/10.1002/ar.20916> | [Google Scholar](#)

Yu H. H., Rosa M. G. P (2014) **Uniformity and diversity of response properties of neurons in the primary visual cortex: Selectivity for orientation, direction of motion, and stimulus size from center to far periphery** *Visual Neuroscience* **31**:85–98 <https://doi.org/10.1017/S0952523813000448> | [Google Scholar](#)

Yu H.-H., Chaplin T. A., Rosa M. G. P (2015) **Representation of central and peripheral vision in the primate cerebral cortex: Insights from studies of the marmoset brain** *Neuroscience Research* **93**:47–61 <https://doi.org/10.1016/j.neures.2014.09.004> | [Google Scholar](#)

Yu H.-H., Rowley D. P., Price N. S. C., Rosa M. G. P., Zavitz E (2020) **A twisted visual field map in the primate dorsomedial cortex predicted by topographic continuity** *Sci Adv* **6** <https://doi.org/10.1126/sciadv.aaz8673> | [Google Scholar](#)

Zhuang J., Ng L., Williams D., Valley M., Li Y., Garrett M., Waters J (2017) **An extended retinotopic map of mouse cortex** *eLife* **6** <https://doi.org/10.7554/eLife.18372> | [Google Scholar](#)

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Reviewer #1 (Public review):

Summary:

In this manuscript, the authors argue that defining higher visual areas (HVAs) based on reversals of retinotopic tuning has led to an over-parcellation of secondary visual cortices. Using retinotopic models, they propose that the HVAs are more parsimoniously mapped as a single area V2, which encircles V1 and exhibits complex retinotopy. They reanalyze functional data to argue that functional differences between HVAs can be explained by retinotopic coverage. Finally, they compare the classification of mouse visual cortex to that of other species to argue that our current classification is inconsistent with those used in other model species.

Strengths:

This manuscript is bold and thought-provoking, and is a must-read for mouse visual neuroscientists. The authors take a strong stance on combining all HVAs, with the possible exception of area POR, into a single V2 region. Although I suspect many in the field will find that their proposal goes too far, many will agree that we need to closely examine the assumptions of previous classifications to derive a more accurate areal map. The authors' supporting analyses are clear and bolster their argument. Finally, they make a compelling argument for why the classification is not just semantic, but has ramifications for the design of experiments and analysis of data.

Weaknesses:

Although I enjoyed the polemic nature of the manuscript, there are a few issues that weaken their argument.

(1) Although the authors make a compelling argument that retinotopic reversals are insufficient to define distinct regions, they are less clear about what would constitute convincing evidence for distinct visual regions. They mention that a distinct area V3 has been (correctly) defined in ferrets based on "cytoarchitecture, anatomy, and functional properties", but elsewhere argue that none of these factors are sufficient to parcellate any of the HVAs in mouse cortex, despite some striking differences between HVAs in each of these factors. It would be helpful to clearly define a set of criteria that could be used for classifying distinct regions.

(2) On a related note, although the authors carry out impressive analyses to show that differences in functional properties between HVAs could be explained by retinotopy, they glossed over some contrary evidence that there are functional differences independent of retinotopy. For example, axon projections to different HVAs originating from a single V1 injection - presumably including neurons with similar retinotopy - exhibit distinct functional properties (Glickfeld LL et al, Nat Neuro, 2013). As another example, interdigitated M2+/M2- patches in V1 show very different HVA connectivity and response properties, again

independent of V1 location/retinotopy (Meier AM et al., bioRxiv). One consideration is that the secondary regions might be considered a single V2 with distinct functional modules based on retinotopy and connectivity (e.g., V2LM, V2PM, etc).

(3) Some of the HVAs-such as AL, AM, and LI-appear to have redundant retinotopic coverage with other HVAS, such as LM and PM. Moreover, these regions have typically been found to have higher "hierarchy scores" based on connectivity (Harris JA et al., Nature, 2019; D'Souza RD et al., Nat Comm, 2022), though unfortunately, the hierarchy levels are not completely consistent between studies. Based on existing evidence, there is a reasonable argument to be made for a hybrid classification, in which some regions (e.g., LM, P, PM, and RL) are combined into a single V2 (though see point #2 above) while other HVAs are maintained as independent visual regions, distinct from V2. I don't expect the authors to revise their viewpoint in any way, but a more nuanced discussion of alternative classifications is warranted.

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Reviewer #2 (Public review):

Summary:

The study by Rowley and Sedigh-Sarvestani presents modeling data suggesting that map reversals in mouse lateral extrastriate visual cortex do not coincide with areal borders, but instead represent borders between subregions within a single area V2. The authors propose that such an organization explains the partial coverage in higher-order areas reported by Zhuang et al., (2017). The scheme revisits an organization proposed by Kaas et al., (1989), who interpreted the multiple projection patches traced from V1 in the squirrel lateral extrastriate cortex as subregions within a single area V2. Kaas et al's interpretation was challenged by Wang and Burkhalter (2007), who used a combination of topographic mapping of V1 connections and receptive field recordings in mice. Their findings supported a different partitioning scheme in which each projection patch mapped a specific topographic location within single areas, each containing a complete representation of the visual field. The area map of mouse visual cortex by Wang and Burkhalter (2007) has been reproduced by hundreds of studies and has been widely accepted as ground truth (CCF) (Wang et al., 2020) of the layout of rodent cortex. In the meantime, topographic mappings in marmoset and tree shrew visual cortex made a strong case for map reversals in lateral extrastriate cortex, which represent borders between functionally diverse subregions within a single area V2. These findings from non-rodent species raised doubts about whether during evolution, different mammalian branches have developed diverse partitioning schemes of the cerebral cortex. Rowley and Sedigh-Sarvestani favor a single master plan in which, across evolution, all mammalian species have used a similar blueprint for subdividing the cortex.

Strengths:

The story illustrates the enduring strength of science in search of definitive answers.

Weaknesses:

To me, it remains an open question whether Rowley and Sedigh-Sarvestani have written the final chapter of the saga. A key reason for my reservation is that the areas the maps used in their model are cherry-picked. The article disregards published complementary maps, which show that the entire visual field is represented in multiple areas (i.e. LM, AL) of lateral extrastriate cortex and that the map reversal between LM and AL coincides precisely with the transition in m2AChR expression and cytoarchitecture (Wang and Burkhalter, 2007; Wang

et al., 2011). Evidence from experiments in rats supports the gist of the findings in the mouse visual cortex (Coogan and Burkhalter, 1993).

(1) The selective use of published evidence, such as the complete visual field representation in higher visual areas of lateral extrastriate cortex (Wang and Burkhalter, 2007; Wang et al., 2011) makes the report more of an opinion piece than an original research article that systematically analyzes the area map of mouse visual cortex we have proposed. No direct evidence is presented for a single area V2 with functionally distinct subregions.

(2) The article misrepresents evidence by commenting that m2AChR expression is mainly associated with the lower field. This is counter to published findings showing that m2AChR spans across the entire visual field (Gamanut et al., 2018; Meier et al., 2021). The utility of markers for delineating areal boundaries is discounted, without any evidence, in disregard of evidence for distinct areal patterns in early development (Wang et al., 2011). Pointing out that markers can be distributed non-uniformly within an area is well-familiar. m2AChR is non-uniformly expressed in mouse V1, LM and LI (Ji et al., 2015; D'Souza et al., 2019; Meier et al., 2021). Recently, it has been found that the patchy organization within V1 plays a role in the organization of thalamocortical and intracortical networks (Meier et al., 2025). m2AChR-positive patches and m2AChR-negative interpatches organize the functionally distinct ventral and dorsal networks, notably without obvious bias for upper and lower parts of the visual field.

(3) The study has adopted an area partitioning scheme, which is said to be based on anatomically defined boundaries of V2 (Zhuang et al., 2017). The only anatomical borders used by Zhuang et al. (2017) are those of V1 and barrel cortex, identified by cytochrome oxidase staining. In reality, the partitioning of the visual cortex was based on field sign maps, which are reproduced from Zhuang et al., (2017) in Figure 1A. It is unclear why the maps shown in Figures 2E and 2F differ from those in Figure 1A. It is possible that this is an oversight. But maintaining consistent areal boundaries across experimental conditions that are referenced to the underlying brain structure is critical for assigning modeled projections to areas or sub-regions. This problem is evident in Figure 2F, which is presented as evidence that the modeling approach recapitulates the tracings shown in Figure 3 of Wang and Burkhalter (2007). The dissimilarities between the modeling and tracing results are striking, unlike what is stated in the legend of Figure 2F.

(4) The Rowley and Sedigh-Sarvestani find that the partial coverage of the visual field in higher order areas shown by Zhuang et al (2017) is recreated by the model. It is important to caution that Zhuang et al's (2017) maps were derived from incomplete mappings of the visual field, which was confined to -25-35 deg of elevation. This underestimates the coverage we have found in LM and AL. Receptive field mappings show that LM covers 0-90 deg of azimuth and -30-80 elevation (Wang and Burkhalter, 2007). AL covers at least 0-90 deg of azimuth and -30-50 deg of elevation (Wang and Burkhalter, 2007; Wang et al., 2011). These are important differences. Partial coverage in LM and AL underestimates the size of these areas and may map two projection patches as inputs to subregions of a single area rather than inputs to two separate areas. Complete, or nearly complete, visual representations in LM and AL support that each is a single area. Importantly, both areas are included in a callosal-free zone (Wang and Burkhalter, 2007). The surrounding callosal connections align with the vertical meridian representation. The single map reversal is marked by a transition in m2AChR expression and cytoarchitecture (Wang et al., 2011).

(5) The statement that the "lack of visual field overlap across areas is suggestive of a lack of hierarchical processing" is predicated on the full acceptance of the mappings by Zhuang et al (2017). Based on the evidence reviewed above, the reclassification of visual areas proposed in Figure 1C seems premature.

(6) The existence of lateral connections is not unique to rodent cortex and has been described in primates (Felleman and Van Essen, 1991).

(7) Why the mouse and rat extrastriate visual cortex differ from those of many other mammals is unclear. One reason may be that mammals with V2 subregions are strongly binocular.

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Reviewer #3 (Public review):

Summary:

The authors review published literature and propose that a visual cortical region in the mouse that is widely considered to contain multiple visual areas should be considered a single visual area.

Strengths:

The authors point out that relatively new data showing reversals of visual-field sign within known, single visual areas of some species require that a visual field sign change by itself should not be considered evidence for a border between visual areas.

Weaknesses:

The existing data are not consistent with the authors' proposal to consolidate multiple mouse areas into a single "V2". This is because the existing definition of a single area is that it cannot have redundant representations of the visual field. The authors ignore this requirement, as well as the data and definitions found in published manuscripts, and make an inaccurate claim that "higher order visual areas in the mouse do not have overlapping representations of the visual field". For quantification of the extent of overlap of representations between 11 mouse visual areas, see Figure 6G of Garrett et al. 2014. [Garrett, M.E., Nauhaus, I., Marshel, J.H., and Callaway, E.M. (2014). Topography and areal organization of mouse visual cortex. The Journal of neuroscience 34, 12587-12600. 10.1523/JNEUROSCI.1124-14.2014.

<https://doi.org/10.7554/eLife.105910.1.sa1>

Author response:

eLife Assessment:

This paper performs a valuable critical reassessment of anatomical and functional data, proposing a reclassification of the mouse visual cortex in which almost all the higher visual areas are consolidated into a single area V2. However, the evidence supporting this unification is incomplete, as the key experimental observations that the model attempts to reproduce do not accurately reflect the literature. This study will likely be of interest to neuroscientists focused on the mouse visual cortex and the evolution of cortical organization.

We do not agree or understand which 'key experimental observations' that the model attempts to reproduce do not accurately reflect the literature. The model reproduces a complete map of the visual field, with overlap in certain regions. When reversals are used to delineate areas, as is the current custom, multiple higher order areas are generated, and each area has a biased and overlapping visual field coverage. These are the simple outputs of the

model, and they are consistent with the published literature, including recent publications such as Garrett et al. 2014 and Zhuang et al. 2017, a paper published in this journal. The area boundaries produced by the model are not identical to area boundaries in the literature, because the model is a simplification.

Reviewer #1 (Public review):

Summary:

In this manuscript, the authors argue that defining higher visual areas (HVAs) based on reversals of retinotopic tuning has led to an over-parcellation of secondary visual cortices. Using retinotopic models, they propose that the HVAs are more parsimoniously mapped as a single area V2, which encircles V1 and exhibits complex retinotopy. They reanalyze functional data to argue that functional differences between HVAs can be explained by retinotopic coverage. Finally, they compare the classification of mouse visual cortex to that of other species to argue that our current classification is inconsistent with those used in other model species.

Strengths:

This manuscript is bold and thought-provoking, and is a must-read for mouse visual neuroscientists. The authors take a strong stance on combining all HVAs, with the possible exception of area POR, into a single V2 region. Although I suspect many in the field will find that their proposal goes too far, many will agree that we need to closely examine the assumptions of previous classifications to derive a more accurate areal map. The authors' supporting analyses are clear and bolster their argument. Finally, they make a compelling argument for why the classification is not just semantic, but has ramifications for the design of experiments and analysis of data.

Weaknesses:

Although I enjoyed the polemic nature of the manuscript, there are a few issues that weaken their argument.

(1) Although the authors make a compelling argument that retinotopic reversals are insufficient to define distinct regions, they are less clear about what would constitute convincing evidence for distinct visual regions. They mention that a distinct area V3 has been (correctly) defined in ferrets based on "cytoarchitecture, anatomy, and functional properties", but elsewhere argue that none of these factors are sufficient to parcellate any of the HVAs in mouse cortex, despite some striking differences between HVAs in each of these factors. It would be helpful to clearly define a set of criteria that could be used for classifying distinct regions.

We agree the revised manuscript would benefit from a clear discussion of updated rules of area delineation in the mouse. In brief, we argue that retinotopy alone should not be used to delineate area boundaries in mice, or any other species. Although there is some evidence for functional property, architecture, and connectivity changes across mouse HVAs, area boundaries continue to be defined primarily, and sometimes solely (Garrett et al., 2014; Juavinett et al., 2018; Zhuang et al., 2017), based on retinotopy. We acknowledge that earlier work (Wang and Burkhalter, 2007; Wang et al., 2011) did consider cytoarchitecture and connectivity alongside retinotopy, but more recent work has shifted to a focus on retinotopy as indicated by the currently accepted criterion for area delineation.

As reviewer #2 points out, the present criteria for mouse visual area delineation can be found in the Methods section of: [Garrett, M.E., Nauhaus, I., Marshel, J.H., and Callaway, E.M. (2014)].

Criterion 1: Each area must contain the same visual field sign at all locations within the area.

Criterion 2: Each visual area cannot have a redundant representation of visual space.

Criterion 3: Adjacent areas of the same visual field sign must have a redundant representation.

Criterion 4: An area's location must be consistently identifiable across experiments.

As discussed in the manuscript, recent evidence in higher order visual cortex of tree shrews and rats led us to question the universality of these criteria across species. Specifically, tree shrew V2, macaque V2, and marmoset DM, exhibit reversals in visual field-sign in what are defined as single visual areas. This suggests that criterion 1 should be updated. It also suggests that Criterion 2 and 3 should be updated since visual field sign reversals often co-occur with retinotopic redundancies, since reversing course in the direction of progression along the visual field can easily lead to coverage of visual field regions already traveled.

More broadly, we argue that topography is just one of several criteria that should be considered in area delineation. We understand that few visual areas in any species meet all criteria, but we emphasize that topography cannot consistently be the sole satisfied criterion – as it currently appears to be for many mouse HVAs. Inspired by a recent perspective on cortical area delineation (Petersen et al., 2024), we suggest the following rules, that will be worked into the revised version of the manuscript. Topography is a criterion, but it comes after considerations of function, architectonics and connectivity.

- (1) Function—Cortical areas differ from neighboring areas in their functional properties
- (2) Architectonics—Cortical areas often exhibit distinctions from neighboring areas in multiple cyto- and myeloarchitectonic markers
- (3) Connectivity—Cortical areas are characterized by a specific set of connectional inputs and outputs from and to other areas
- (4) Topography—Cortical areas often exhibit a distinct topography that balances maximal coverage of the sensory field with minimal redundancy of coverage within an area.

As we discuss in the manuscript, although there are functional, architectonic, and connectivity differences across mouse HVAs, they typically vary smoothly across multiple areas – such that neighboring areas share the same properties and there are no sharp borders. For instance, sharp borders in cytoarchitecture are generally lacking in the mouse HVAs. A notable exceptions to this is the clear and sharp change in m2AChR expression that occurs between LM and AL (Wang et al., 2011).

(2) On a related note, although the authors carry out impressive analyses to show that differences in functional properties between HVAs could be explained by retinotopy, they glossed over some contrary evidence that there are functional differences independent of retinotopy. For example, axon projections to different HVAs originating from a single V1 injection - presumably including neurons with similar retinotopy - exhibit distinct functional properties (Glickfeld LL et al, Nat Neuro, 2013). As another example, interdigitated M2+/M2- patches in V1 show very different HVA connectivity and response properties, again independent of V1 location/retinotopy (Meier AM et al., bioRxiv). One consideration is that the secondary regions might be considered a single V2 with distinct functional modules based on retinotopy and connectivity (e.g., V2LM, V2PM, etc).

Thank you for the correction. We will revise the text to discuss (Glickfeld et al., 2013), as it remains some of the strongest evidence in favor of retinotopy-independent functional specialization of mouse HVAs. However, one caveat of this study is the size of the V1 injection that is the source of axons studied in the HVAs. As apparent in Figure 1B, the large injection

covers nearly a quarter of V1. It is worth noting that (Han et al., 2018) found, using single-cell reconstructions and MAPseq, that the majority of V1 neurons project to multiple nearby HVA targets. In this experiment the tracing does not suffer from the problem of spreading over V1's retinotopic map, and suggests that, presumably retinotopically matched, locations in each area receive shared inputs from the V1 population rather than a distinct but spatially interspersed subset. In fact, the authors conclude "Interestingly, the location of the cell body within V1 was predictive of projection target for some recipient areas (Extended Data Fig. 8). Given the retinotopic organization of V1, this suggests that visual information from different parts of visual field may be preferentially distributed to specific target areas, which is consistent with recent findings (Zhuang et al., 2017)". Given an injection covering a large portion of the retinotopic map, and the fact that feed-forward projections from V1 to HVAs carry coarse retinotopy - it is difficult to prove that functional specializations noted in the HVA axons are retinotopy-independent. This would require measurement of receptive field location in the axonal boutons, which the authors did not perform (possibly because the SNR of calcium indicators prevented such measurements at the time).

Another option would be to show that adjacent neurons in V1, that project to far-apart HVAs, exhibit distinct functional properties on par with differences exhibited by neurons in very different parts of V1 due to retinotopy. In other words, the functional specificity of V1 inputs to HVAs at retinotopically identical locations is of the same order as those that might be gained by retinotopic biases. To our knowledge, such a study has not been conducted, so we have decided to measure the data in collaboration with the Allen Institute. As part of the Allen Institute's pioneering OpenScope project, we will make careful two-photon and electrophysiology measurements of functional properties, including receptive field location, SF, and TF in different parts of the V1 retinotopic map. Pairing this data with existing Allen Institute datasets on functional properties of neurons in the HVAs will allow us to rule in, or rule-out, our hypotheses regarding retinotopy as the source of functional specialization in mouse HVAs. We will update the discussion in the revised manuscript to better reflect the need for additional evidence to support or refute our proposal.

Meier AM et al., bioRxiv 2025 (Meier et al., 2025) was published after our submission, but we are thankful to the reviewers for guiding our attention to this timely paper. Given the recent findings on the influence of locomotion on rodent and primate visual cortex, it is very exciting to see clearly specialized circuits for processing self-generated visual motion in V1. However, it is difficult to rule out the role of retinotopy as the HVA areas (LM, AL, RL) participating in the M2+ network less responsive to self-generated visual motion exhibit a bias for the medial portion of the visual field and the HVA area (PM) involved in the M2- network responsive to self-generated visual motion exhibit a bias for the lateral (or peripheral) parts of the visual field. For instance, a peripheral bias in area PM has been shown using retrograde tracing as in Figure 6 of (Morimoto et al., 2021), single-cell anterograde tracing as in Extended Data Figure 8 of (Han et al., 2018), and functional imaging studies (Zhuang et al., 2017). Recent findings in the marmoset also point to visual circuits in the peripheral, but not central, visual field being significantly modulated by self-generated movements (Rowley et al., 2024).

However, a visual field bias in area PM that selectively receive M2- inputs is at odds with the clear presence of modular M2+/M2- patches across the entire map of V1 (Ji et al., 2015). One possibility supported by existing data is that neurons in M2- patches, as well as those in M2+ patches, in the central representation of V1 make fewer or significantly weaker connections with area PM compared to areas LM, AL and RL. Evidence to the contrary would support retinotopy-independent and functionally specialized inputs from V1 to HVAs.

(3) Some of the HVAs-such as AL, AM, and LI-appear to have redundant retinotopic coverage with other HVAs, such as LM and PM. Moreover, these regions have typically been found to have higher "hierarchy scores" based on connectivity (Harris JA et al.,

Nature, 2019; D'Souza RD et al., Nat Comm, 2022), though unfortunately, the hierarchy levels are not completely consistent between studies. Based on existing evidence, there is a reasonable argument to be made for a hybrid classification, in which some regions (e.g., LM, P, PM, and RL) are combined into a single V2 (though see point #2 above) while other HVAs are maintained as independent visual regions, distinct from V2. I don't expect the authors to revise their viewpoint in any way, but a more nuanced discussion of alternative classifications is warranted.

We understand that such a proposal would combine a subset of areas with matched field sign (LM, P, PM, and RL) would be less extreme and received better by the community. This would create a V2 with a smooth map without reversals or significant redundant retinotopic coverage. However, the intuition we have built from our modeling studies suggest that both these areas, and the other smaller areas with negative field sign (AL, AM, LI), are a byproduct of a complex single map of the visual field that exhibits reversals as it contorts around the triangular and tear-shaped boundaries of V1. In other words, we believe the redundant coverage and field-sign changes/reversals are a byproduct of a single secondary visual field in V2 constrained by the cortical dimensions of V1. That being said, we understand that area delineations are in part based on a consensus by the community. Therefore we will continue to discuss our proposal with community members, and we will incorporate new evidence supporting or refuting our hypothesis, before we submit our revised manuscript.

Reviewer #2 (Public review):

Summary:

The study by Rowley and Sedigh-Sarvestani presents modeling data suggesting that map reversals in mouse lateral extrastriate visual cortex do not coincide with areal borders, but instead represent borders between subregions within a single area V2. The authors propose that such an organization explains the partial coverage in higher-order areas reported by Zhuang et al., (2017). The scheme revisits an organization proposed by Kaas et al., (1989), who interpreted the multiple projection patches traced from V1 in the squirrel lateral extrastriate cortex as subregions within a single area V2. Kaas et al's interpretation was challenged by Wang and Burkhalter (2007), who used a combination of topographic mapping of V1 connections and receptive field recordings in mice. Their findings supported a different partitioning scheme in which each projection patch mapped a specific topographic location within single areas, each containing a complete representation of the visual field. The area map of mouse visual cortex by Wang and Burkhalter (2007) has been reproduced by hundreds of studies and has been widely accepted as ground truth (CCF) (Wang et al., 2020) of the layout of rodent cortex. In the meantime, topographic mappings in marmoset and tree shrew visual cortex made a strong case for map reversals in lateral extrastriate cortex, which represent borders between functionally diverse subregions within a single area V2. These findings from non-rodent species raised doubts about whether during evolution, different mammalian branches have developed diverse partitioning schemes of the cerebral cortex. Rowley and Sedigh-Sarvestani favor a single master plan in which, across evolution, all mammalian species have used a similar blueprint for subdividing the cortex.

Strengths:

The story illustrates the enduring strength of science in search of definitive answers.

Weaknesses:

To me, it remains an open question whether Rowley and Sedigh-Sarvestani have written the final chapter of the saga. A key reason for my reservation is that the areas the maps used in their model are cherry-picked. The article disregards published complementary

maps, which show that the entire visual field is represented in multiple areas (i.e. LM, AL) of lateral extrastriate cortex and that the map reversal between LM and AL coincides precisely with the transition in m2AChR expression and cytoarchitecture (Wang and Burkhalter, 2007; Wang et al., 2011). Evidence from experiments in rats supports the gist of the findings in the mouse visual cortex (Coogan and Burkhalter, 1993).

We would not claim to have written the final chapter of the saga. Our goal was to add an important piece of new evidence to the discussion of area delineations across species. We believe this new evidence supports our unification hypothesis. We also believe that there are several missing pieces of data that could support or refute our hypothesis. We have begun a collaboration to collect some of this data.

(1) The selective use of published evidence, such as the complete visual field representation in higher visual areas of lateral extrastriate cortex (Wang and Burkhalter, 2007; Wang et al., 2011) makes the report more of an opinion piece than an original research article that systematically analyzes the area map of mouse visual cortex we have proposed. No direct evidence is presented for a single area V2 with functionally distinct subregions.

This brings up a nuanced issue regarding visual field coverage. Wang & Burkhalter, 2007 Figure 6 shows the receptive field of sample neurons in area LM that cover the full range between 0 and 90 degrees of azimuth, and -40 to 80 degree of elevation – which essentially matches the visual field coverage in V1. However, we do not know whether these neurons are representative of most neurons in area LM. In other words, while these single-cell recordings along selected contours in cortex show the span of the visual field coverage, they may not be able to capture crucial information about its shape, missing regions of the visual field or potential bias. To mitigate this, visual field maps measured with electrophysiology are commonly produced by even sampling across the two dimensions of the visual area, either by moving a single electrode along a grid-pattern (e.g. (Manger et al., 2002)), or using a grid-linked multi-electrode probe (e.g. (Yu et al., 2020)). This was not carried out either in Wang & Burkhalter 2007 or Wang et al. 2011. Even sampling of cortical space is time consuming and difficult with electrophysiology, but efficient with functional imaging. Therefore, despite the likely under-estimation of visual field coverage, imaging techniques are valuable in that they can efficiently exhibit not only the span of the visual field of a cortical region, but also its shape and bias.

Multiple functional imaging studies that simultaneously measure visual field coverage in V1 and HVAs report a bias in the coverage of HVAs, relative to that in V1 (Garrett et al., 2014; Juavinett et al., 2018; Zhuang et al., 2017). While functional imaging will likely underestimate receptive fields compared to electrophysiology, the consistent observation of an orderly bias for distinct parts of the visual field across the HVAs suggests that at least some of the HVAs do not have full and uniform coverage of the visual field comparable to that in V1. For instance, (Garrett et al., 2014) show that the total coverage in HVAs, when compared to V1, is typically less than half (Figure 6D) and often irregularly shaped.

Careful measurements of single-cell receptive fields, using mesoscopic two-photon imaging across the HVAs would settle this question. As reviewer #1 points out, this is technically feasible, though no dataset of this kind exists to our knowledge.

(2) The article misrepresents evidence by commenting that m2AChR expression is mainly associated with the lower field. This is counter to published findings showing that m2AChR spans across the entire visual field (Gamanut et al., 2018; Meier et al., 2021). The utility of markers for delineating areal boundaries is discounted, without any evidence, in disregard of evidence for distinct areal patterns in early development (Wang et al., 2011). Pointing out that markers can be distributed non-uniformly within an area is well-

familiar. m2AChR is non-uniformly expressed in mouse V1, LM and LI (Ji et al., 2015; D'Souza et al., 2019; Meier et al., 2021). Recently, it has been found that the patchy organization within V1 plays a role in the organization of thalamocortical and intracortical networks (Meier et al., 2025). m2AChR-positive patches and m2AChR-negative interpatches organize the functionally distinct ventral and dorsal networks, notably without obvious bias for upper and lower parts of the visual field.

We wrote that “Future work showed boundaries in labeling of histological markers such as SMI-32 and m2ChR labeling, but such changes mostly delineated area LM/AL (Wang et al., 2011) and seemed to be correlated with the representation of the lower visual field.” The latter statement regarding the representation of the lower visual field is directly referencing the data in Figure 1 of (Wang et al., 2011), which is titled “Figure 1: LM/AL border identified by the transition of m2AChR expression coincides with receptive field recordings from lower visual field.” Similar to the Wang et al., we were simply referring to the fact that the border of area LM/AL co-exhibits a change in m2AChR expression as well as lower-visual field representation.

(3) The study has adopted an area partitioning scheme, which is said to be based on anatomically defined boundaries of V2 (Zhuang et al., 2017). The only anatomical borders used by Zhuang et al. (2017) are those of V1 and barrel cortex, identified by cytochrome oxidase staining. In reality, the partitioning of the visual cortex was based on field sign maps, which are reproduced from Zhuang et al., (2017) in Figure 1A. It is unclear why the maps shown in Figures 2E and 2F differ from those in Figure 1A. It is possible that this is an oversight. But maintaining consistent areal boundaries across experimental conditions that are referenced to the underlying brain structure is critical for assigning modeled projections to areas or sub-regions. This problem is evident in Figure 2F, which is presented as evidence that the modeling approach recapitulates the tracings shown in Figure 3 of Wang and Burkhalter (2007). The dissimilarities between the modeling and tracing results are striking, unlike what is stated in the legend of Figure 2F.

Thanks for this correction. By “anatomical boundaries of higher visual cortex”, we meant the cortical boundary between V1 and higher order visual areas on one end, and the outer edge of the envelope that defines the functional boundaries of the HVAs in cortical space (Zhuang et al., 2017). The reviewer is correct that we should have referred to these as functional boundaries. The word ‘anatomical’ was meant to refer to cortical space, rather than visual field space.

More generally though, there is no disagreement between the partitioning of visual cortex in Figure 1 and 2. Rather, the portioning in Figure 1 is directly taken from Zhuang et al., (2017) whereas those in Figure 2 are produced by mathematical model simulation. As such, one would not expect identical areal boundaries between Figure 2 and Figure 1. What we aimed to communicate with our modeling results, is that a single area can exhibit multiple visual field reversals and retinotopic redundancies if it is constrained to fit around V1 and cover a visual field approximately matched to the visual field coverage in V1. We defined this area explicitly as a single area with a single visual field (boundaries shown in Figure 2A). So the point of our simulation is to show that even an explicitly defined single area can appear as multiple areas if it is constrained by the shape of mouse V1, and if visual field reversals are used to indicate areal boundaries. As in most models, different initial conditions and parameters produce a complex visual field which will appear as multiple HVAs when delineated by areal boundaries. What is consistent however, is the existence of complex single visual field that appears as multiple HVAs with partially overlapping coverage.

Similarly, we would not expect a simple model to exactly reproduce the multi-color tracer injections in Wang and Burkhalter (2007). However, we find it quite compelling that the

model can produce multiple groups of multi-colored axonal projections beyond V1 that can appear as multiple areas each with their own map of the visual field using current criteria, when the model is explicitly designed to map a single visual field. We will explain the results of the model, and their implications, better in the revised manuscript.

(4) The Rowley and Sedigh-Sarvestani find that the partial coverage of the visual field in higher order areas shown by Zhuang et al (2017) is recreated by the model. It is important to caution that Zhuang et al's (2017) maps were derived from incomplete mappings of the visual field, which was confined to -25-35 deg of elevation. This underestimates the coverage we have found in LM and AL. Receptive field mappings show that LM covers 0-90 deg of azimuth and -30-80 elevation (Wang and Burkhalter, 2007). AL covers at least 0-90 deg of azimuth and -30-50 deg of elevation (Wang and Burkhalter, 2007; Wang et al., 2011). These are important differences. Partial coverage in LM and AL underestimates the size of these areas and may map two projection patches as inputs to subregions of a single area rather than inputs to two separate areas. Complete, or nearly complete, visual representations in LM and AL support that each is a single area. Importantly, both areas are included in a callosal-free zone (Wang and Burkhalter, 2007). The surrounding callosal connections align with the vertical meridian representation. The single map reversal is marked by a transition in m2AChR expression and cytoarchitecture (Wang et al., 2011).

This is a good point. We do not expect that expanding the coverage of V1 will change the results of the model significantly. However, for the revised manuscript, we will update V1 coverage to be accurate, repeat our simulations, and report the results.

(5) The statement that the "lack of visual field overlap across areas is suggestive of a lack of hierarchical processing" is predicated on the full acceptance of the mappings by Zhuang et al (2017). Based on the evidence reviewed above, the reclassification of visual areas proposed in Figure 1C seems premature.

The reviewer is correct. In the revised manuscript, we will be careful to distinguish bias in visual field coverage across areas from presence or lack of visual field overlap.

(6) The existence of lateral connections is not unique to rodent cortex and has been described in primates (Felleman and Van Essen, 1991).

(7) Why the mouse and rat extrastriate visual cortex differ from those of many other mammals is unclear. One reason may be that mammals with V2 subregions are strongly binocular.

This is an interesting suggestion, and careful visual topography data from rabbits and other lateral eyed animals would help to evaluate it. For what it's worth, tree shrews are lateral eyed animals with only 50 degrees of binocular visual field and also show V2 subregions.

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by itself should not be considered evidence for a border between visual areas.

Weaknesses:

The existing data are not consistent with the authors' proposal to consolidate multiple mouse areas into a single "V2". This is because the existing definition of a single area is that it cannot have redundant representations of the visual field. The authors ignore this requirement, as well as the data and definitions found in published manuscripts, and make an inaccurate claim that "higher order visual areas in the mouse do not have overlapping representations of the visual field". For quantification of the extent of overlap of representations between 11 mouse visual areas, see Figure 6G of Garrett et al. 2014. [Garrett, M.E., Nauhaus, I., Marshel, J.H., and Callaway, E.M. (2014). Topography and areal organization of mouse visual cortex. The Journal of neuroscience 34, 12587-12600. 10.1523/JNEUROSCI.1124-14.2014.

Thank you for this correction, we admit we should have chosen our words more carefully. In the revised manuscript, we will emphasize that higher order visual areas in the mouse do have some overlap in their representations but also exhibit bias in their coverage. This is consistent with our proposal and in fact our model simulations in Figure 2E also show overlapping representations along with differential bias in coverage. However, we also note Figure 6 of Garret et al. 2014 provides several pieces of evidence in support of our proposal that higher order areas are sub-regions of a single area V2. Specifically, the visual field coverage of each area is significantly less than that in V1 (Garret et al. 2014, Figure 6D). While the imaging methods used in Garret et al. likely under-estimate receptive fields, one would assume they would similarly impact measurements of coverage in V1 and HVAs. Secondly, each area exhibits a bias towards a different part of the visual field (Figure 6C and E), that this bias is distinct for different areas but proceeds in a retinotopic manner around V1 - with adjacent areas exhibiting biases for nearby regions of the visual field (Figure 6E). Thus, the biases in the visual field coverage across HVAs appear to be related and not independent of each other. As we show in our modeling and in Figure 2, such orderly and inter-related biases can be created from a single visual field constrained to share a border with mouse V1.

With regards to the existing definition of a single area: we did not ignore the requirement that single areas cannot have redundant representations of the visual field. Rather, we believe that this requirement should be relaxed considering new evidence collected from other species, where multiple visual field reversals exist within the same visual area. We understand this issue is nuanced and was not made clear in the original submission.

In the revised manuscript, we will clarify that visual field reversals often exhibit redundant retinotopic representation on either side of the reversal. In the revised manuscript we will clarify that our argument that multiple reversals can exist within a single visual area in the mouse, is an argument that some retinotopic redundancy can exist with single visual areas. Such a re-classification would align how we define visual areas in mice with existing classification in tree shrews, ferrets, cats, and primates – all of whom have secondary visual areas with complex retinotopic maps exhibiting multiple reversals and redundant retinotopic coverage.

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